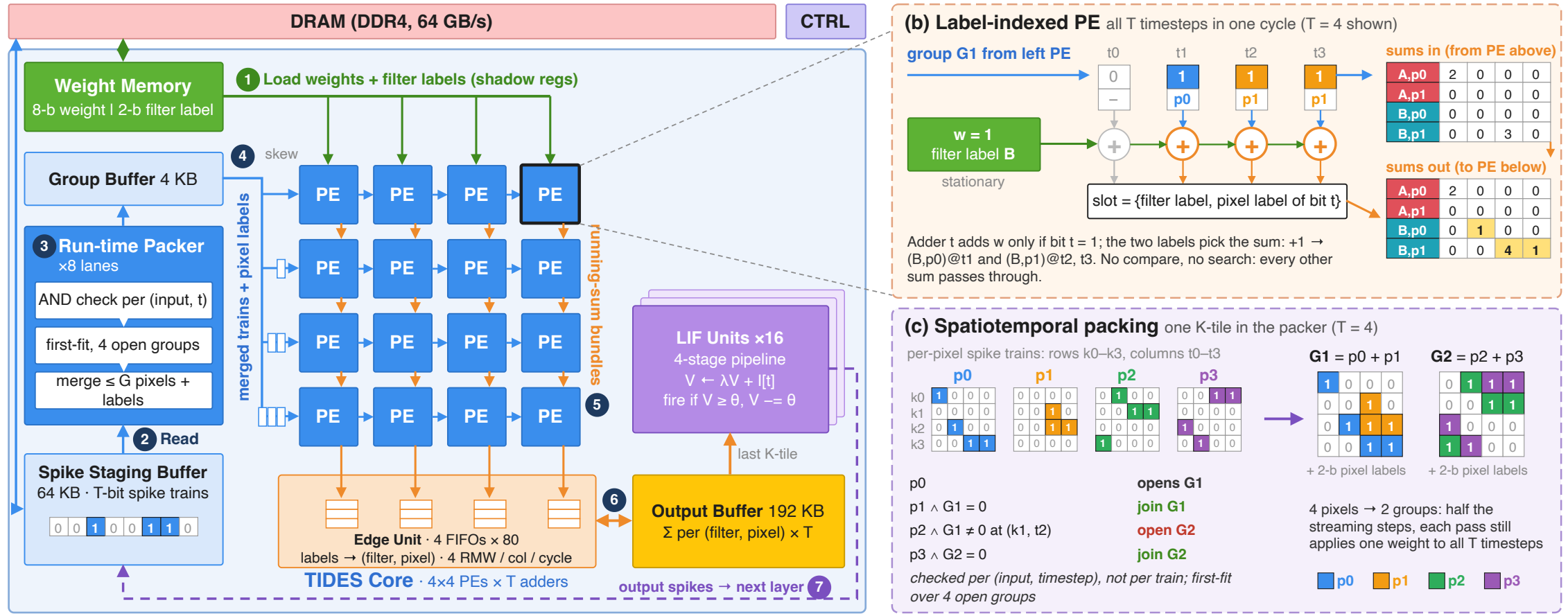

DRAM (DDR4, 64 GB/s)
CTRL
Weight Memory
8-b weight | 2-b filter label
1 Load weights + filter labels (shadow regs)
Group Buffer 4 KB
4 skew
3 Run-time Packer
×8 lanes
AND check per (input, t)
first-fit, 4 open groups
merge ≤ G pixels + labels
2 Read
Spike Staging Buffer
64 KB · T-bit spike trains
merged trains + pixel labels
PE
running-sum bundles
5
LIF Units ×16
4-stage pipeline
V ← λV + I[t]
fire if V ≥ θ, V −= θ
last K-tile
Edge Unit · 4 FIFOs × 80
labels → (filter, pixel) · 4 RMW / col / cycle
6
Output Buffer 192 KB
Σ per (filter, pixel) × T
TIDES Core · 4×4 PEs × T adders
output spikes → next layer
7
(b) Label-indexed PE all T timesteps in one cycle (T = 4 shown)
group G1 from left PE
t0
t1
t2
t3
w = 1
filter label B
stationary
slot = {filter label, pixel label of bit t}
sums in (from PE above)
sums out (to PE below)
A,p0
A,p1
B,p0
B,p1
Adder t adds w only if bit t = 1; the two labels pick the sum: +1 → (B,p0)@t1 and (B,p1)@t2, t3. No compare, no search: every other sum passes through.
(c) Spatiotemporal packing one K-tile in the packer (T = 4)
per-pixel spike trains: rows k0–k3, columns t0–t3
p0
p1
p2
p3
k0
k1
k2
k3
G1 = p0 + p1
G2 = p2 + p3
+ 2-b pixel labels
+ 2-b pixel labels
p0 opens G1
p1 ∧ G1 = 0 join G1
p2 ∧ G1 ≠ 0 at (k1, t2) open G2
p3 ∧ G2 = 0 join G2
checked per (input, timestep), not per train; first-fit over 4 open groups
4 pixels → 2 groups: half the streaming steps, each pass still applies one weight to all T timesteps
p0
p1
p2
p3